

BILLEL ZIGHAB

Senior Solution Architect

Enterprise & Solution Architecture · Cloud / On-Premise / Hybrid · Standards, Governance & Assurance

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English · French · Arabic — Fluent

PROFESSIONAL SUMMARY

Senior Solution Architect, 17+ years of IT experience including 10+ years of end-to-end architecture on critical, high-volume digital platforms. Designs large-scale distributed and transactional architectures and defines applicative and cloud evolution trajectories (target + transition) in international contexts. Specialist in microservices architectures, event-driven decoupling (EDA, CQRS) and API Design / API Management strategies (gateway, contracts, versioning). Drives the urbanization of complex systems and supports teams on cross-system integration choices, challenging technical decisions and making the right performance / cost / time trade-offs. Owns application security by design: authentication, authorization and token management (OAuth2/OIDC, JWT, mTLS), IAM / CIAM, Zero Trust. Strong command of AWS and working knowledge of GCP, hybrid / multi-cloud architectures, high availability, resilience and data integrity. Also designs the integration architecture for AI/ML capabilities developed by data science teams. Recognized for explaining complex topics to business and technical stakeholders and uniting teams in Agile / multicultural settings.

TECHNICAL SKILLS

Frameworks, Governance & Architecture Deliverables	TOGAF, ArchiMate, TAD (Technical Architecture Document), HLD, LLD, ADR, C4 models, flow & impact matrices, DR/BCP plan, architecture diagrams, data model, reference patterns, API contracts (OpenAPI)...
AI/ML — Architecture & Integration	Integration architecture for AI/ML capabilities within the enterprise IS. GenAI on Azure AI Foundry: Azure OpenAI (GPT-4o), RAG / Azure AI Search, embeddings (text-embedding-3-large), AI Gateway (APIM GenAI mode), Content Safety. Computer Vision (CNN, EfficientNet), TensorFlow/Keras. Interface contracts, data & security governance, feasibility ADRs.
Application & Data Architecture Patterns	Microservices, EDA, CQRS/Event Sourcing, Saga, API Gateway & BFF, Strangler Fig, Data Mesh, Lakehouse, Lambda/Kappa Architecture, DDD, Polyglot Persistence
Cloud (Azure / AWS)	AKS, APIM, Databricks, Synapse, Event Hubs, Functions, Key Vault, Front Door, WAF, ADLS Gen2, Cosmos DB · EKS, Lambda, MSK, SageMaker, API GW, S3, Redshift · Terraform, Bicep, Helm, Kubernetes, Docker
Security & Compliance	Zero Trust, Defense in Depth, IAM/Federation, mTLS, PKI, Key Management, OWASP, GDPR, IRM, SAML, JWT, PKCE, OAuth2/OIDC
Modeling	BiZZdesign Enterprise Studio, SAP PowerDesigner, ArchiMate, C4/Structurizr, Visio, Lucidchart
Dev & DevSecOps	Java/Spring Boot, Python, Scala, REST/OpenAPI, Angular, React · GitLab CI, Jenkins, SonarQube, Checkmarx, Artifactory
Data	Oracle, PostgreSQL, SQL Server, MongoDB, Cassandra, Elasticsearch, Cloudera/HDP (Spark, HDFS, Hive)

PROFESSIONAL EXPERIENCE

Shell — Trading & Supply NBS · Carbon Management · Power Portfolio

Feb 2022 – Present

Senior Solution Architect · Greater London, UK · Hybrid

- Framed, designed and governed the end-to-end architecture of 3 mission-critical portfolios across 30 multi-region Azure applications, from business-requirements capture to production, with a 99.9%+ SLA and a DR plan (RTO < 1h / RPO < 15 min).
- Designed a real-time event-driven architecture (Spark Structured Streaming, CQRS, Saga) for Flex GB battery trading, reducing Day-Ahead decision latency by 40%.
- Designed and governed an enterprise GenAI platform on Azure AI Foundry: a knowledge assistant (RAG) for trading and operations teams, grounded on internal references (procedures, contracts, market notes), served by GPT-4o with vector search (text-embedding-3-large embeddings).

- Set the production foundations: AI Gateway (APIM GenAI mode) — per-application token quotas, PTU routing with PAYG overflow, content-safety policies — with keyless access (Entra ID, managed identities, private endpoints, managed VNet) and cost control (PTU/PAYG trade-offs).
- Formalized the architecture documentation chain (conceptual/logical/technical views, HLD, LLD, ADR) defended in ARB and Design Authority, cutting approval time by 35% for 8 teams.
- Reduced cloud costs by 25% (shared landing zones, AKS/Databricks scaling) while enforcing at-rest/in-transit encryption (TLS/mTLS, Key Vault).
- Led cost-optimization initiatives with the SOM and Enterprise Architecture teams.

Deliverables : TAD, HLD, LLD, ADR, flow & impact matrices, DR/BCP plan.

Tech stack : Azure AI Foundry, Azure OpenAI (GPT-4o, text-embedding-3-large), RAG / Azure AI Search, AI Gateway (APIM GenAI mode), Content Safety, Entra ID & managed identities, Private Endpoints, AKS, Databricks, Spark Structured Streaming, Event Hubs, Cosmos DB, Key Vault, Terraform, Bicep.

ENEDIS — Département Data (Chaîne de valeur Données / B4ALL)

Jul 2020 – Feb 2022

Senior Solution Architect · Paris, France · Agile / SAFe

- Designed the cloud-native architecture of the Linky meter quality-control platform (28M) integrating a Computer Vision model (EfficientNet-B4) built by the data science team: AWS exposure/integration, interface contracts, security — feasibility and integrity guaranteed (94% F1-score).
- Led end-to-end architectural design, governance and modernization of the Data Department, aligned with group standards (Open-Source First, Cloud First).
- Defined the inter-application interface architecture (API contracts, protocols) and conceptual/technical views, qualifying data sensitivity with business & security.
- Modernized 7 legacy IS components (technical debt ~30%); created 12 reusable patterns adopted by 5 teams (time-to-design ~30%).
- Ran open-source benchmarks in the architecture committee (licensing costs ~40%).

Deliverables : TAD, flow matrices, data model, reference patterns, ADR.

Tech stack : Cloudera/HDP, Kafka, AWS (SageMaker, Lambda, API GW, S3), TensorFlow/Keras, Python, Spring Boot, Kubernetes, Docker.

Europcar Mobility Group — Click & Go

Jan 2019 – Jun 2020

Solution Architect · Paris, France · Agile / Scrum

- Full architectural audit of the platform (B2B & B2C portal + mobile app): 15 technical-debt points identified and a 4-quarter refactoring roadmap validated by technical management.
- Architecture built around a centralized API Gateway (Kong) and microservices decomposition, with explicit interface contracts and TLS encryption — cutting inter-service coupling by about 60% and time-to-release by about 45%.
- Cross-cutting security architecture (Spring Security, LDAP, OAuth2) for partner and public authentication, GDPR-compliant.
- Designed solution components and architecture models for the Spring Batch (bulk reservation processing) and Spring Data JPA (Oracle/PostgreSQL access) layers, producing HLD/LLD deliverables, technical analyses and ADRs for governance reviews.
- CI/CD pipeline (Jenkins, SonarQube, Artifactory, GitLab, Newman) reducing regressions by about 45% through test automation.

Deliverables : TAD, HLD, LLD, ADR, API contracts (OpenAPI), security document, technical analyses.

Tech stack : Java 11, Spring Boot 2.x, Spring Security, Spring Batch, Spring MVC REST, Spring LDAP, Spring Data JPA, Microservices, Kong API Gateway, Oracle, PostgreSQL, Business Objects, Android & iOS, Docker, OpenShift, Angular 7, Bootstrap, Jenkins, SonarQube, Artifactory, GitLab, Newman, IntelliJ.

RTE — Réseau de Transport d'Électricité

May 2018 – Jan 2019

Technical Solution Architect · Paris, France · Agile / Scrum

- Established the reference technical architecture of the GUIDE platform (50,000+ external identities) — Spring Boot + OpenLDAP + Angular — and authored the architecture vision documents (application, global, functional), delivered on time with 0 critical defects in production.
- Trained and coached 12 developers in DDD, TDD, CI and design-pattern practices, reducing the defect rate by 40% within the first 3 sprints post-coaching.

Deliverables : Architecture vision documents (application, global, functional), TAD, HLD/LLD.

Tech stack : Java 8, Spring Boot 2.x, Spring Security, Spring LDAP, REST APIs, Angular 6, Docker, Jenkins, SonarQube, OpenLDAP, Maven 3.

AFPA

Jan 2017 – Apr 2018

Application & Technical Architect / Technical Lead · Paris, France · Scrum

- Maintenance and evolution of the existing modules within the CIRUS scope, ensuring service continuity and consistency with the existing architecture.
- Full rebuild of selected modules — including GESPERS — redesigned from scratch on recent technologies (microservices/API, Angular, security, observability).
- Definition of the conceptual and technical architecture of the modules involved and production of the deliverables (HLD/LLD, ADR), presented to and validated by the enterprise architecture committee.
- Support to the development team as technical lead, from design through to production, over a 16-month engagement.

Tech stack : Java 8, Spring Boot, Spring Security, Hibernate, Docker, Angular 4, SQL Server 2012, MySQL, Jenkins, SonarQube, GitLab.

ENEDIS — OpenData

Apr 2016 – Dec 2016

Big Data Technical Lead · Paris, France · Scrum

- Designed and delivered the OpenData large-scale data publication and processing pipeline (Spark, Hadoop), processing 30 public datasets with 99% availability.
- Coached 6 junior developers in Big Data practices and Git branching, reducing the integration error rate by 60% over the last 2 sprints.

Tech stack : Java 8, Spark, Hadoop, Hive, HDFS, Docker, Teradata, Jenkins, SonarQube.

Airbus Helicopters — ORION

Jul 2013 – Mar 2016

Technical Lead & Junior Technical Architect · *Marseille, France · Agile / Scrum*

- Harmonized and industrialized 18 application components on the ORION platform, reducing code-duplication debt by 35% and improving the maintainability of documentation and maintenance manuals.
- Developed an API health-check system reducing MTTR (Mean Time To Recovery) by 60% by detecting service failures before user impact. Deployed full DevOps automation (Jenkins, Nexus, Sonar, Git).

Tech stack : Java/JEE, REST, AngularJS, Angular 2, XML/XSLT/XPath, Spring Boot, Docker, Jenkins, Nexus, SonarQube, SQL Server, MySQL, Active Directory LDAP, IntelliJ, Maven 3.

SNCF — Portail Affilié CPRP

Oct 2011 – Jun 2013

Senior Java/JEE Developer · *Marseille, France · Agile*

- Designed and developed the complete architecture of the CPRP affiliate portal (Java/JEE, WebLogic, Oracle 11g, REST, Liferay 6.2) in 8 months, delivered on time with 7,000 affiliated users migrated without service interruption.

Tech stack : Java/JEE, EJB 3, Liferay 6.2, Hibernate, JSF, REST (RESTEasy), Oracle 11g, Maven, Tomcat 7, Git.

Direction Générale des Finances Publiques

Feb 2011 – Sep 2011

Talend & Senior Java Developer · *Paris, France · Agile*

- Developed and delivered the note-sharing and financial-control system in 7 months. Successfully executed the MySQL → PostgreSQL and Drupal Commons 6 → 7 data migration with 0 data loss across 600,000 records.

Tech stack : Java 6, Talend Studio, Hibernate, Spring, Maven 2, PrimeFaces, MySQL, PostgreSQL.

Arts et Métiers ParisTech

Jul 2009 – Jan 2011

Java/JEE Developer · *France · Agile*

- Delivered the critical business interfaces between the KAKI payroll module and the Concerto Adix accounting system, ensuring zero-defect integration of payroll for 1,200 employees into the ENSAM financial system.

Tech stack : Java/JEE, Spring MVC, Spring Security, Oracle 10g, JasperSoft, Apache POI, PL/SQL, JBoss, Maven, Ant, Selenium.

Ville de Toulon & Métropole TPM

Jul 2008 – Jun 2009

PHP Developer · *Toulon, France · Agile*

- Rebuilt and delivered the toulon.fr & tpm-agglo.fr websites on Drupal 7 — 8 custom modules developed, 120K visitors/month supported after go-live.

Tech stack : Drupal 7, PHP, HTML5, CSS, jQuery, Apache 2, MySQL, Solr, Git.

Université Aix-Marseille — Saint-Jérôme Faculty Library

Jan 2008 – Jun 2008

Java/JEE Developer · *Aix-Marseille, France · Agile*

- Rebuilt the library's document management system (Java/JEE, Spring, Hibernate, SOAP), delivered and deployed across 3 environments (test, training, production) within 6 months.

Tech stack : Java/JEE, Spring, Hibernate, JSF, PrimeFaces, Tomcat 6, SOAP/WSDL, MySQL, JasperReports, SVN.

EDUCATION

Master's in Distributed Systems

2008

Université Aix-Marseille, Aix-Marseille, France

LEADERSHIP & DISTINCTIONS

- Mentored 8+ senior architects and engineers at Shell, contributed to the group Technology Radar and led cloud cost-optimization initiatives — a cross-cutting role recognized beyond the direct project scope.
- Evangelized modern engineering practices (DDD, TDD, CI, Zero Trust, Security by Design) across 12+ development teams at RTE, AFPA and Airbus Helicopters over 3 consecutive assignments.
- Trilingual English / French / Arabic (fluent) — facilitator of collaboration in international and cross-cultural multi-team contexts (Shell UK, EDF, Europcar, DGFIP).
- Presented and defended 50+ architecture proposals in governance bodies (Design Authority, architecture committees), notably Shell EA Board, ENEDIS CONSUEL, RTE Architecture Board.